

Shanghai Wander: Vibe & Vision

Part 1. Abstract

This project proposes an "City Walk Route Guide" designed for young tourists with limited leisure time who focus on short trips, enjoy city landscapes and entertainment consumption. The project will initially focus on five famous "city-walk" streets in Shanghai: Yuyuan Road, Anfu Road, Wukang Road, People's Square and Huaihai Road. By organizing multiple visual data (photos, descriptions, and social media comments) on interactive Google Maps (Fig.1), it provides users more referenceability to explore cities and enjoy short vacations, avoiding the unpleasant experience of wasting limited time. As tool for developing multi-dimensional information and assisting urban exploration, this project will promote public awareness and reflection on "internet celebrity culture" and improving SDG 12[7], cater to the consumption and travel habits of young people, and provide an example of information visualization for other tourism industry.

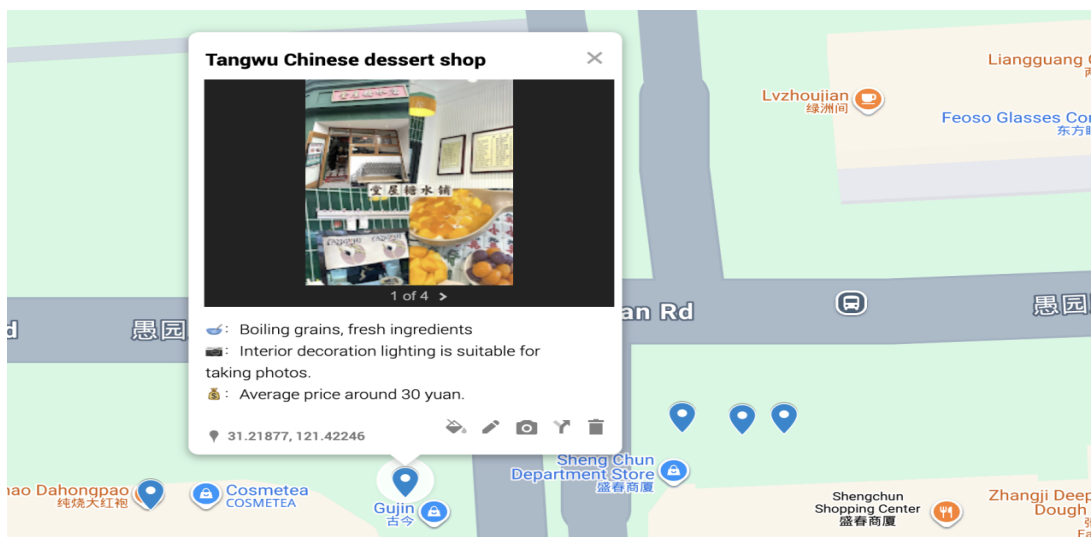


Fig. 1 Teaser Figure: Interactive Map Interface Showing City Walk Routes with Filters and User Reviews.

Academic Professionalism Block

Disclaimer

This project is submitted for INFOSCI 301 Data Visualization and Information Aesthetics, Spring 2026, instructed by Professor Luyao Zhang at Duke Kunshan University

Acknowledgments

We thank Professor Luyao Zhang for her guidance, our peers for valuable feedback, and our community partners—Jiading Urban Planning Exhibition Hall, Waving Cube Museum, and Kunshan Museum of Chinese Opera—for inspiring our design.

Contribution to SDG Goals

- SDG 11 (Sustainable Cities): Encourages balanced urban exploration and reduces over-tourism.
- SDG 12 (Responsible Consumption): Helps users make informed choices, supporting authentic local businesses.

Statement of Accessibility

All materials are publicly available:

GitHub: <https://github.com/FionaChen623/City-Wander-with-Vibe-Vision>

Hugging Face: https://huggingface.co/datasets/JocelynGUO/INFO_Shanghai_Map_Guide

5. Keywords

Information visualization; city walk; community-based learning; sustainable tourism; interactive map; Shanghai; cultural heritage; responsible consumption; bilingual accessibility.

Part 2. Value — Community-Based Learning & Social Impact



Fig. 2 Community Impact Flow Chart

2.1 Targeted Community

Our visualization project mainly serves young tourists and residents in Shanghai who have limited leisure time and focus on budget. This group is the most active participant and subject in the city's "internet celebrity" consumer culture, relying on social media to explore travel destinations. However, they are now facing serious information asymmetry, often encountering disappointing experiences such as idealized online performance and high prices (such as 50 yuan per cup of coffee), overly commercialized spaces, and real-life scenes that are completely different from social media images. Our project enables this community to break through the dilemma of information asymmetry, develop urban landscape tours through multidimensional information, and make full use of limited rest time and financial resources. To serve Shanghai's international population, the interface supports both Chinese and English—main map elements are bilingual, while the public transport module (relying on domestic APIs) remains in Chinese for accuracy.

2.2 Community based learning connections

Our project has a different emphasis on viewing content compared to field trips. The Jiading Urban Planning Exhibition Hall showcases the top-down, data-driven urban development process and future vision, which is an official narrative presentation of data. In contrast, our project uses data visualization to build a bottom-up, ordinary citizen based platform (as shown in Fig.2). This project utilizes visualization technology to help users penetrate the consumer fantasy woven by algorithms and capital, and touch on a more grounded experiential reality. Therefore, our map integrates these insights and uses data visualization technology to critically reveal the "deceptive" internet celebrity tourism economy, while also providing comprehensive information for young tourists with corresponding needs, so that they can weigh the pros and cons and make better tourism route planning with abundant information.

2.3 Applied Learning & Critical Reflection

The platform serves as an applied learning tool that explicitly designed around the critical reflection cycle—experience, analysis, abstraction, and application—which is fundamental to deepening learning in community-based contexts [4]. And it will transform them into practical resources for community self-protection and sustainable consumption.

Firstly, our project will implement the key findings of Jakarta spatial research. This study suggests that the aesthetic of "installable programs" is not an organic cultural trend, but a

thoughtful business strategy aimed at redesigning spaces to maximize foot traffic and revenue [1]. There is a direct negative inference in the consumption pattern of Shanghai: it promotes both homogenization (streets crowded with visual similarity and theme driven bakeries and cafes) and deceptive marketing (a large number of edited photos masking mediocre reality). Our map addresses this issue by crowdsourcing and visualizing the gap between business planning images and life experiences, enabling users seeing through strategies and make informed choices.

Secondly, our methodology is inspired by the practicality of architectural level spatial analysis in Hong Kong [3]. Like Hong Kong, Shanghai is a high-density mega city with extremely scarce and expensive land resources. This economic pressure exacerbates the problems as businesses face high rents that are often passed on to consumers at inflated prices. In this situation we must promote the effective utilization of existing urban structures. Therefore, by highlighting regions and companies that provide true value, authenticity, and fair pricing, we guide users to make reasonable choices.

By linking the diagnosis of the problem with the method of finding solutions, our project transforms users from passive urban spatial consumption to active and analytical understanding of their economic and social forces, ultimately forming more powerful and sustainable consumption behaviors.

2.4 SDG Alignment

This project focuses on the unhealthy "internet-celebrity economy" and related consumption, which will help reduce unnecessary transportation related carbon dioxide emissions and resource waste (such as food waste and time waste) and contribute to SDG 11 sustainable cities and SDG 12 responsible consumption.[7]

Part3. Data Sources & Data Governance

3.1 Data source

To ensure the depth, reliability, and public accountability of the project, our data work integrates authoritative open data with carefully curated community-generated content and divided into 3 clear stages:

- (1) Official Open Data (Shanghai Municipal Administration of Culture and

Tourism): We will use the official open dataset released by the Shanghai Culture and Tourism Bureau which provides verified information on public facilities, and cultural sites along targeted routes. It serves as a reliable baseline for mapping and helps us avoid factual errors in venue location and basic classification.

- (2) Core data (manually collected by designers): This is the cornerstone of the initial version of the project. We implement a systematic data collection process for stores along the target routes: Firstly, information filtering(collected from social media, excluded official marketing content from merchants), content analysis (store attributes, space assessment and consumption level), and future expansion (user crowdsourcing).

Data governance

We adhere to the FAIR principles (Findable, Accessible, Interoperable, Reusable) and the CARE principles (Collective Benefit, Authority to Control, Responsibility, Ethics) to ensure data transparency and trustworthiness. For social media data, all user identifiers are removed, photo EXIF data is stripped, and only publicly accessible posts are collected in accordance with platform terms of service. Datasets are fully documented with metadata schemas (Table.1 and Fig 3) and hosted on Hugging Face for reproducibility

Table 1. Data metadata table

Dataset Name	Source	Description	Variables	Temporal Coverage	License	Privacy Handling
Shanghai Cultural Open Data	Shanghai Municipal Administration of Culture and Tourism	Verified public facilities and cultural sites	Name, address, type, coordinates	2023–2024	Open data	No privacy issues
Social Media Comments	Xiaohongshu, Dianping	User reviews and photos	Rating, comment text, tags, image URLs	2024–2025	Educational use only	Anonymized, no user IDs

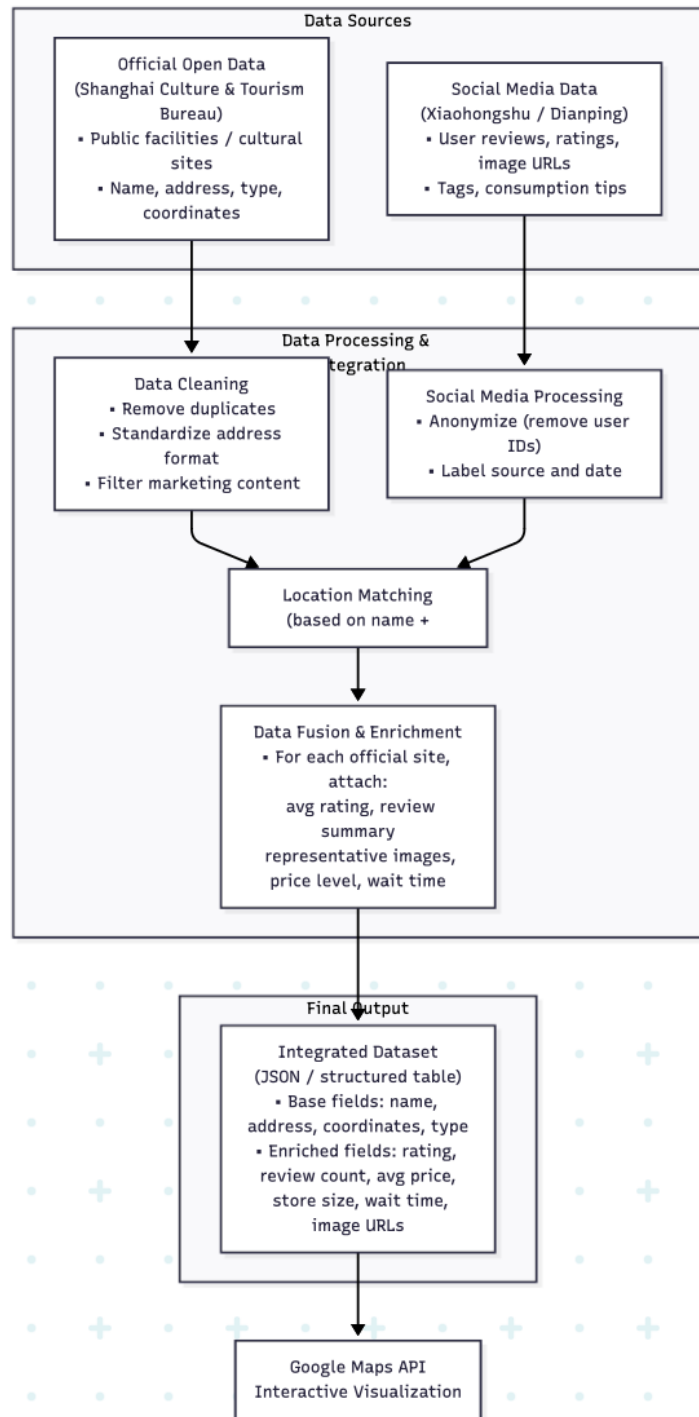


Fig. 3 Integrating official data with user-generated content
(Complete preprocessing script can be found on GitHub)

Part 4. Visualization Tools & Design Flow

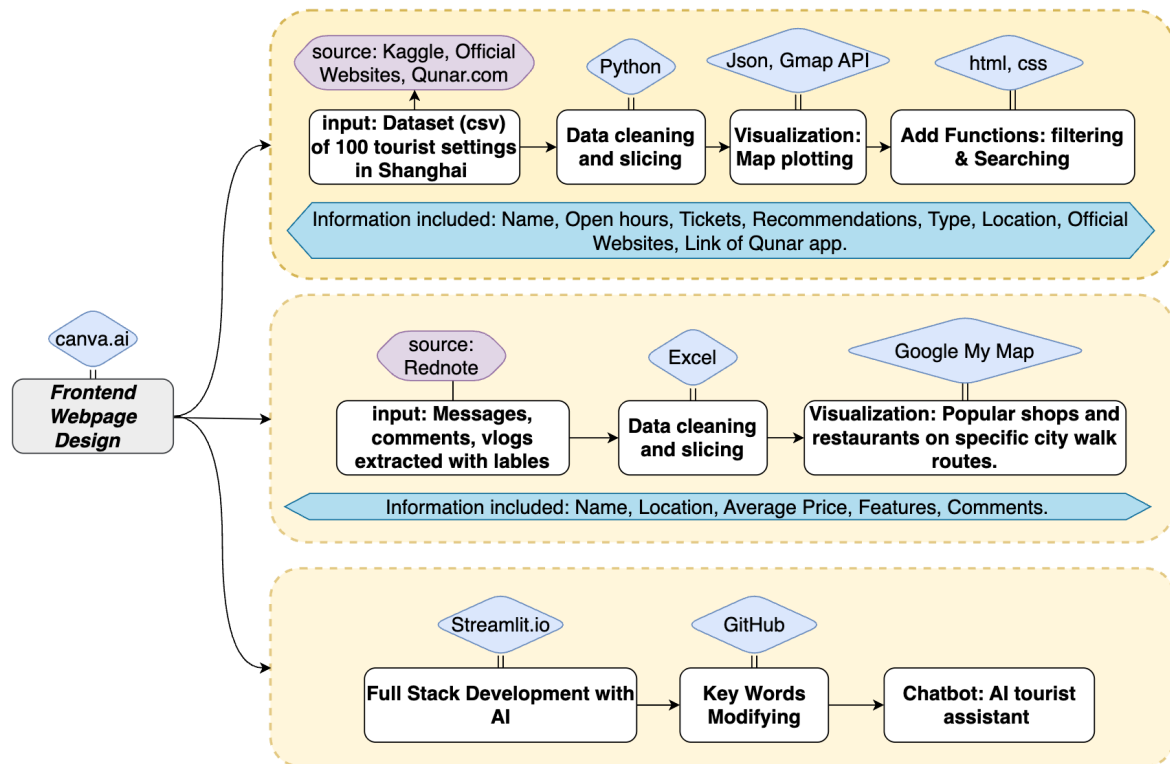


Fig. 4: Updated Design Flow

4.1 Tools & Platforms

Aligned with the course's focus on interactive, community-engaged visualizations, our stack prioritizes accessibility and dynamic user interaction. We use Python (Pandas) for cleaning structured tourist datasets from Kaggle and Qunar, while Excel processes unstructured community data from Rednote. Geospatial visualization relies on the Google Maps, Json and Gmap API, paired with HTML, CSS, and JavaScript for the client-side interface. An AI chatbot is built with Streamlit.io and hosted on GitHub, enabling conversational, personalized data access.

4.2 Updated Design Flow

As shown in Fig. 4, the webpage built with canva contains three main functions, each flowchart is illustrated respectively in the figure.

(1) Data Pipeline: The pipeline starts with two inputs: a CSV of 100 Shanghai tourist settings and labeled Rednote community content. Python cleans the tourist data, while Excel processes community attributes like price and comments, with updates synced to the web interface. Cleaned data visualizes as map markers (Google Maps API) and route overlays (Google My Map), with the Streamlit chatbot offering conversational access. JavaScript

powers interactions: filtering markers, zooming, clicking for InfoWindows, and chatbot queries. The project structure of the first function is listed in Fig. 5, and the right schema displays the cleaned bilingual data with variables.

(2) Key User Pathways: Users begin by exploring the map, hovering for previews and zooming to spot clusters. They deepen engagement by clicking markers for community-driven content or using sidebar filters to narrow results by category or price. The chatbot lets them adjust keywords for tailored suggestions, turning passive viewing into active storytelling.

(3) Expected Moments of Insight: Users discover hidden community hubs via aggregated reviews, identify spatial patterns like Xuhui’s green space clusters through filtered maps, and connect cultural spots to personal interests via the chatbot, fostering a deeper understanding of Shanghai’s urban culture.

shanghai-photo-map/ ├─ index.html ├─ README.md ├─ data/ │ ├── shanghai_spots.xlsx │ └── places.json ├─ scripts/ │ └── excel_to_json.py ├─ images/ │ ├── wukang/ │ └── bund/ ├─ modules/ │ └── photo_guide.html ├─ css/ │ └── style.css ├─ js/ │ ├── main.js │ ├── map.js │ └── filter.js └─ .gitignore (optional) (optional)	Column	Type	Description
	id	text	Unique identifier for each location
	name	text	Name of the place
	lat	number	Latitude
	lng	number	Longitude
	tags	text	Tags (comma-separated, e.g. street,architecture,night)
	images	text	Image paths (comma-separated, relative paths, e.g. images/wukang/1.jpg)
	description	text	Short description of the location
	tips	text	Photography tips or shooting suggestions

Fig. 5: Project framework and CSV data schema

Left: web application structure (HTML, CSS, JS, Python scripts, images, modules). Right: Excel schema defining columns feeding JSON for interactive visualization.

Part 5. Research Inspiration & Innovation

5.1 Academic & Real-World Inspirations

Our project is informed by two key academic works and multiple real-world projects. We draw on ReVISit 2’s interactive visualization framework from IEEE VIS 2025 [5], which supports structured user interaction and systematic data management for user-centered visual analytics. We also build on Wu et al. [6], who use social media data to reveal city-scale urban behavioral patterns. In practice, we reference *The New York Times’ “Mapping America”* for narrative-driven geographic visualization. We further examine the “沪小游” WeChat mini

program (Fig. 6), which provides official route overviews but lacks user-generated content, price information, and timely updates. We also identify limitations in the international version of Meituan, which provides fragmented information and overemphasizes commercial content rather than curated city walk experiences.



Fig. 6 Interface of the “沪小游 Huxiaoyou” WeChat mini program.

5.2 Innovation Narrative

As demonstrated in Fig. 7, our system extends existing research and addresses critical gaps in current tourism visualization. Unlike ReVISit 2 and Wu et al.’s macro-level analysis, our design focuses on micro-level, route-based urban exploration suitable for young residents and visitors. Compared with “沪小游 Huxiaoyou”, which only provides official descriptions and limited locations, our platform integrates user reviews, price levels, category filtering, and travel links in one interface. Compared with Meituan’s international version, our visualization organizes points along classic city walk routes rather than scattered commercial spots, making it more practical and accessible for international users. These improvements represent meaningful design innovation rather than incremental changes. We also strengthen the evaluation plan through structured user testing and feedback collection. By combining curated routes, multi-source information, and user-friendly interaction, our project contributes to more sustainable, ethical, and community-centered urban visualization.

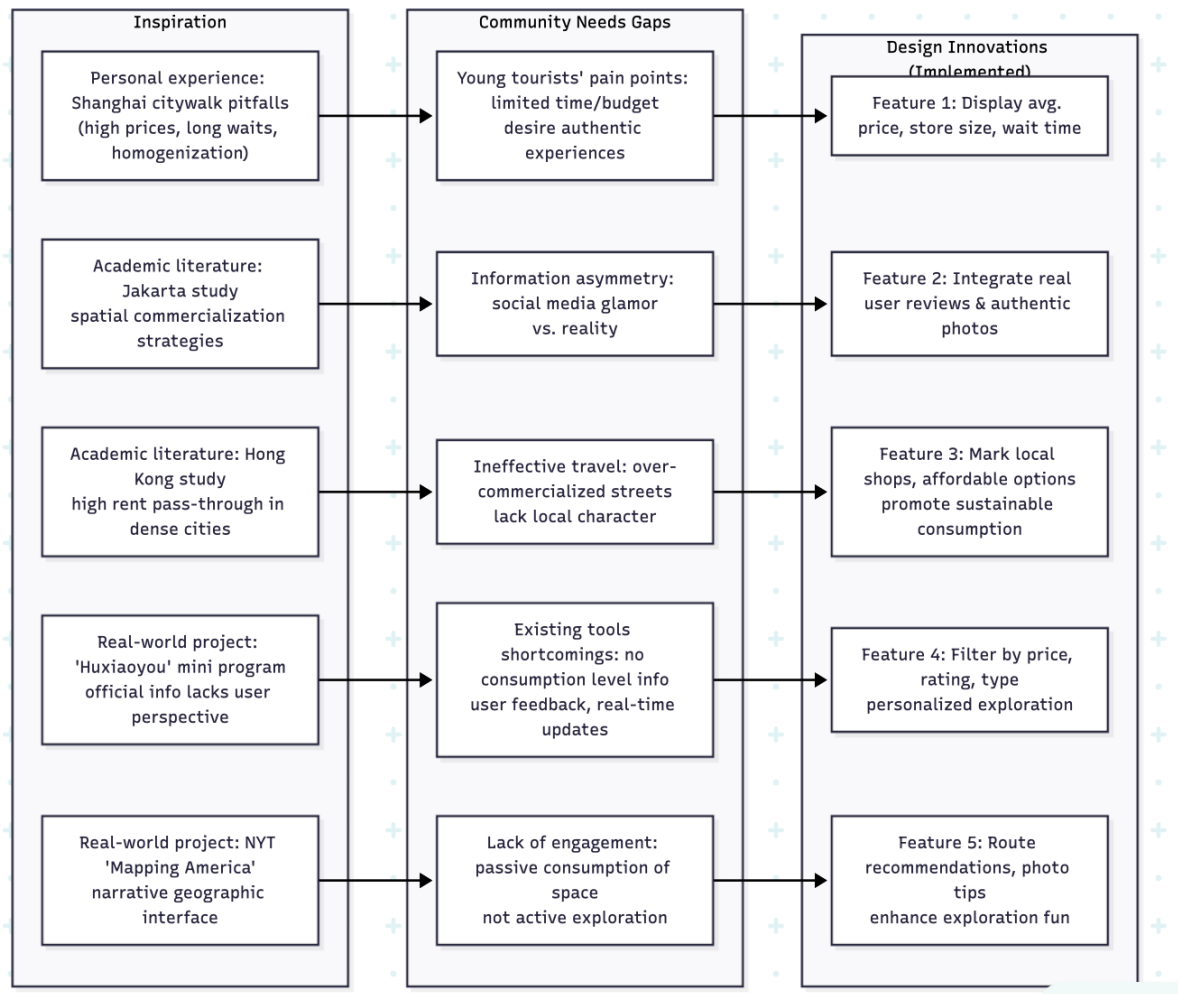


Fig. 7 Research Inspiration Flowchart

Part 6. Team Makeup & Strengths

A. Basic Author Information

Table 2 Author Basic Information

Team Member	Yufei Chen	Yuyang Guo
Major	Computer Science	Data Science
Class / Year	Class of 2028	
University	Duke Kunshan University	

Individual Strengths Relevant to the Project	Python-based data processing, visualization construction, basic web development (HTML, CSS, JSON), technical implementation of interactive features, data cleaning, and tool integration.	Community-oriented research, urban cultural analysis, narrative framing, literature review, cultural context integration, SDG alignment, ethical framing, and user-centered content design.
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Our team consists of two members with complementary skills in data analysis, visualization design, and cultural research. Yuyang, a Data Science student with strong social science training, focuses on community-oriented research, urban cultural analysis, and narrative framing. Yufei, a Computer Science student, contributes technical expertise in Python-based data processing, visualization construction, and basic web development (HTML, CSS, JSON). Both members' prior InfoVis Redesign Projects strengthened their ability to translate complex data and urban issues into accessible visual forms, enabling effective collaboration across research, design, and implementation.

B. Individual Reflection

Yufei Chen

a) Contribution from Individual InfoVis Redesign Project

My individual InfoVis Redesign Project, focused on the Jiading Urban Planning Exhibition Hall, significantly shaped my approach to this team project. The individual work emphasized translating complex urban data into accessible, actionable visualizations for diverse audiences, this is an idea I carried into our city walk visualization. Specifically, I applied the interaction design and data preparation methods from my individual project, such as prioritizing clear user flows and ensuring technical accuracy in data representation. Additionally, the skills I developed in building user-friendly visualization tools (e.g., intuitive navigation and information hierarchy) directly informed my role in implementing charts, interaction flows, and Python-based visualization tools for the team project.

b) Individual Perspective in Community-Based Learning (CBL)

As a Computer Science student, my perspective centers on bridging technical

implementation with community needs, which aligns with the CBL focus of the project. My work supports civic engagement by developing interactive map features (e.g., route guidance, clear marker interactions) that make Shanghai's urban culture and community resources accessible to young residents and tourists. Insights from the Urban Futures field trip reinforced the importance of connectivity and accessibility in city layouts, guiding me to design tools that prioritize user experience and equitable access to information. The Living Cultural Heritage trip further inspired me to integrate local context into technical design, such as descriptive annotations and photos that convey neighborhood character. This work connects to broader learning goals of cultural understanding and sustainability, as the visualization fosters appreciation for urban heritage and supports informed, responsible exploration of the city.

Yuyang Guo

a) Contribution from Individual InfoVis Redesign Project

My individual InfoVis Redesign Project, focused on the Kunshan bus online platform, taught me critical lessons about data readability and user-centered design that I brought to the team project. From my individual work, I learned that excessive text and pure image introductions hinder user understanding—so I proposed presenting store introductions in bullet-point format and using intuitive emojis as key prompts for our city walk visualization. Additionally, my individual project highlighted the importance of spatial accuracy in map-based tools; I applied this by ensuring that the relative positions of storefronts and landmarks in our dataset align with real-world locations, preventing erroneous information. These ideas and methods directly improved the clarity and usability of our team's visualization.

b) Individual Perspective in Community-Based Learning (CBL)

With a background in Data Science and social science training, my perspective focuses on centering community needs and cultural context in the project, making it a meaningful CBL deliverable. My work supports public understanding by integrating local cultural insights and user-generated content into the visualization, helping residents and tourists connect with Shanghai's neighborhoods on a deeper level. The group discussion experience at the Kunqu Opera Museum made me more aware of balancing commercial and cultural values, guiding our team to align the project with SDG 12 (Responsible

Consumption and Production) by highlighting sustainable, community-focused spots rather than overemphasizing commercial locations. This project connects to broader learning goals of cultural understanding and global challenges, as it fosters appreciation for urban heritage and promotes responsible, community-engaged exploration.

C. Team Contribution Statement

Table. 3 Contribution Category

Contribution Category	Team Member Contributions	
	Yuyang Guo	Yufei Chen
Data Assets Contribution	Collected field trip docs (photos, notes); sourced social media data, external references, and cultural context materials.	Handled dataset preparation, cleaning, and validation; ensured data readiness, technical accuracy and consistency for visualization.
Literature & Research Contribution	Primary responsibility for literature material review, cultural context analysis, SDG alignment, and ethical framing.	Conceptual discussions, and translation of research insights into technical requirements.
Visualization & Design Contribution	Designed interactive map concept, spatial visualization framework, and user-centered content (bullet points, emojis)	Developed charts, interaction flows, Visualization tools(API) implemented web interactive features (HTML/CSS/JSON).
Integration & Collaboration	Collaboratively integrated visuals into prototype and refined user experience; balanced cultural depth, technical functionality and accessibility; held regular discussions to reach consensus and ensure equitable responsibility distribution.	

Table 3 summarizes the detailed division of team responsibilities, reflecting transparent authorship and interdisciplinary collaboration in line with academic standards.

Appendix A: Dual-Perspective Assessment

(1) Scientific Community Perspective / Ethos & Logos

Research Contribution:

Our project targets young travelers and residents who are interested in city strolls and are often misled by false information on commercial platforms. Unlike recommendation systems that suggest consumption, our visualization is an information center: aggregates and visualizes from multiple dimensions (price, real photos, user reviews) to help users make informed and autonomous decisions. We address the data transparency gap by manually collecting and publishing verifiable datasets to combat non-open, commercially influenced data from commercial platforms such as Dianping and Xiaohongshu. Additionally, local attractions is visualized on the map through Amap APT, and functions such as search, filtering, and classification can be performed.

Methodological rigor:

Methodology integrates three data sources: Shanghai Open Data (official) and manually curated social media content (anonymized, timestamped). All data is publicly available on Hugging Face with full documentation, ensuring reproducibility (FAIR and CARE principles). The technical pipeline follows a transparent workflow: Python preprocessing → JSON → Google Maps JavaScript API, with frontend built in HTML/CSS and API integration. This design lowers cognitive load by enabling users to filter, compare, and decide within minutes. The approach is theoretically grounded in the Jakarta study [1], which critiques "instagrammable" commercial aesthetics.

(2) User/Community Perspective / Pathos

Target users: Young domestic and international travelers with limited leisure time and budget, who are interested in city walking but often misled by social media information.

Enabled tasks: Users can filter ads by price, view real user photos, read real reviews, and compare options in a bilingual interface. In addition, scenic spots can be classified and searched based on interests

Specific benefits:

- Avoiding excessive hype and misleading leads to waste of time and money.
- Save money by finding affordable local stores.
- Discover communities that align with their interests.

Community value: This platform amplifies real local businesses, reduces the concentration of tourists in a few popular tourist attractions, and supports more balanced urban exploration.

More inspiring video exchange: This video showcases the process of users from opening a map to making informed decisions, demonstrating how the tool addresses the real issues of information overload and distrust.

(3) Synthesis

Strength: Our project delivers a youth-centered, international interactive travel guide that integrates and structures fragmented information from social media, creating a unified, user-friendly resource for exploring Shanghai's urban walking routes.

Limitation: Data collection relies heavily on manual processing; a systematic, automated, and legally compliant bilingual database has not yet been fully established, which restricts real-time updates and long-term scalability.

Improvement: Future work will focus on building a standardized, maintainable database, strengthening collaboration with tourism authorities and international youth communities, and exploring sustainable commercial models to further improve interactivity, usability, and social impact.

Appendix B: Research-Theme Positioning via Think-Pair-Share

(1) Theme Definition

Interdisciplinarity denotes the integration of methods, perspectives, and forms of expertise from multiple disciplines to collaboratively address complex research problems that cannot be adequately understood within a single disciplinary framework.

(2) Scholarly Anchor

This theme is grounded in recent visualization scholarship emphasizing visualization as an enabling infrastructure for interdisciplinary collaboration [8] and is pedagogically framed through the Think–Pair–Share active learning model, which supports structured peer reasoning and conceptual refinement through dialogue [9].

(3) Impact Statement (InfoVis Scholarship)

Interdisciplinarity advances Information Visualization by repositioning visualization as a shared analytical language rather than a purely representational outcome. By enabling the

integration of heterogeneous data and perspectives, interdisciplinary visualization supports collaborative sense-making, enhances interpretability across domains, and broadens the epistemic role of InfoVis in complex, real-world problem solving.

(4) Cross-Team Evaluation

Our project demonstrates interdisciplinarity through a clear division between technical foundations and contextual integration. Data Visualization and Computer Science form the core disciplines, supporting data processing, interactive visualization design, and full-stack development of a web-based city walk guide. These technical components are meaningfully extended through Urban Planning, Social Science, and Cultural Studies. Urban spatial analysis informs the geographic structure of the system, while community-oriented research and consumer culture perspectives shape the narrative and interpretive layers. This integration allows the visualization to function as both a technical system and a socially reflective information tool.

As an exemplar of the interdisciplinary theme, we award the Interdisciplinarity Award to the air quality visualization project by Yuan Yao, Yanzhen Liu, and Zhang Chen. Grounded in environmental science, the project restructures long-term air quality data into interpretable temporal patterns, revealing trends across a decade of environmental change. By linking scientific analysis with public health concerns and the UN Sustainable Development Goals, the project effectively bridges data science, environmental research, and societal impact.

(5) Refinement Narrative (Think–Pair–Share Reflection)

The Think–Pair–Share process played a critical role in refining our disciplinary positioning. Individual reflection clarified our initial assumptions, while paired discussion exposed the limitations of treating interdisciplinarity as merely additive. Group sharing further emphasized visualization’s role as a mediating space for shared reasoning. Through this process, we reframed our project to foreground integration and collaboration rather than parallel disciplinary contributions.

Appendix C: Partnership Dialogue

(1) What do we need from community partners

The main problem facing our project is the lack of commercial data. Consumer platforms, such as Small Red Notes, contains valuable consumer’s experiences, but their anti-

crawling mechanisms make data collection very difficult. And many data is paid advertising rather than real user feedback, which is the problem our project aims to help. To address this problem, we seek the following helps from Kunshan Museum:

Table.4 Required Information and Data

Need	Description
User Data	Visitor demographics, motivations, and behavior patterns to understand which young audiences engage with cultural institutions
Cultural Content	Opera / Kunqu-related materials suitable for young audiences (e.g., stories behind artifacts, food elements in Kunqu, cultural and creative products)
Commercial Activity Information	Future museum initiatives (themed dining, workshops, merchandise) to integrate into our platform as "museum experiences"
Cultural Sensitivity Guidance	Advice on balancing promotion with respect for cultural heritage (deep cooperation with government)

(2) What can we offer to community partners?

The targeted community of our platform are young people interested in "internet celebrity" culture and city walking—students and young professionals with limited leisure time and budgets. This is precisely the audience museums struggle to attract.

Table. 5 Platform Contribution for Partners

Offer	Description
Platform Promotion	Dedicated "Kunshan Opera Museum" section on our interactive map,

	introducing exhibitions and experiences to young users
Targeted Audience Reach	Direct access to young consumers interested in urban exploration.
Commercial Integration	Museum's cultural and creative products and events featured alongside city-walk recommendations
User Insights	Data on how young people interact with cultural content on digital platforms
Student Community	Access to DKU students for user testing and feedback collection

(3) Co-Cooperation Plan

Table. 6 Co-Cooperation Plan

Timeline	Action
Short-term (Spring 2026)	Add a dedicated "Kunshan Museum" section to our interactive map, including basic information, exhibition highlights, cultural artifacts, and visitor tips. This serves as an initial entry point for young users to discover the museum.
Medium-term (2026–2027)	Co-design "Young People's Museum Check-in Guide" including: Photo spots in museum Student ambassador program “Kunqu for Beginners” Series Activities Fashion-tag-promotion on social media
Long-term (2027–2028)	As the geographical scope of the project expands from Shanghai to the wider Jiangsu region, we will integrate Kunshan Museum and nearby cultural landmarks such as Tinglin Park into our interactive map. The platform's classification system (commercial and cultural heritage sites) and

	scrolling feature will promote these contents, making the museum consistently visible to our target audience of young urban explorers.
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(4) Partnership Vision

This collaborative relationship creates common value: our platform enables museums to reach young consumers; Museums provide cultural depth and institutional credibility for our platform. We will work together to transform 'museum visits' from an insufficiently attractive cultural promotion to a natural component of young people's urban exploration. This reciprocal relationship embodies a transformative science of community engagement, where benefits flow in both directions (Pandya et al., 2025).

REFERENCES

- [1] R. D. Aprilian and I. Widiastuti, "The story of adaptive reuse in Jakarta's old building under the 'instagrammable' era," in Proc. 8th Biannual Int. Conf. ARTEPOLIS (ARTEPOLIS 2020), Bandung, Indonesia (Online), 2021. [Online]. Available: <https://doi.org/10.2991/assehr.k.211126.019>
- [2] T. L. Trolan and E. A. Jach, "Applied learning in higher education: Conclusions and recommendations for institutional leadership," *New Directions for Higher Education*, vol. 2019, no. 188, pp. 101–106, 2019.
- [3] A. Maheshwari, Y. Jain, and S. S. Chopra, "Spatial inequity index for essential facility access at building-level in highly dense urban areas," *Cities*, vol. 149, p. 104926, Jul. 2024. [Online]. Available: <https://doi.org/10.1016/j.cities.2024.104926>
- [4] S. L. Ash and P. H. Clayton, "Generating, deepening, and documenting learning: The power of critical reflection in applied learning," *Journal of Applied Learning in Higher Education*, vol. 1, pp. 25–48, Fall 2009.
- [5] Z. Cutler, J. Wilburn, H. Shrestha, Y. Ding, B. Bollen, K. A. Nadib, T. He, A. McNutt, L. Harrison, and A. Lex, "ReVISit 2: A full experiment life cycle user study framework," in Proc. IEEE VIS 2025, Best Full Papers, Vienna, Austria, Nov. 2025. [Online]. Available: https://ieevis.org/year/2025/program/paper_ebdd5b65-b5cb-4d4b-bd78-92ff8cc2f243.html

- [6] H. Wu, S. Zhao, X. Chen, and Y. Wu, “Visual analysis of urban mobility and social media data,” in Proc. IEEE VIS Workshop on Visual Analytics in Science and Technology (VAST), 2019. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6359619/>
- [7] United Nations, The Sustainable Development Goals Report 2023, United Nations, New York, NY, USA, 2023. [Online]. Available: <https://unstats.un.org/sdgs/report/2023/>
- [8] D. Aurich and A. Horaniet Ibañez, “How can data visualization support interdisciplinary research? LuxTIME: studying historical exposomics in Belval,” Frontiers in Big Data, vol. 6, Art. no. 1164885, 2023.
- [9] E. J. Theobald et al., “Active learning narrows achievement gaps for underrepresented students in undergraduate science, technology, engineering, and math,” Proc. Natl. Acad. Sci. U.S.A., vol. 117, no. 12, pp. 6476–6483, 2020.

Project Collaboration Links

-Collaborative Canva Poster URL

https://www.canva.cn/design/DAHAKIaG7JE/WDLXYKaW4u6ePYHpPes9fA/edit?utm_content=DAHAKIaG7JE&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

-Collaborative GitHub Organization or Repository URL

<https://github.com/FionaChen623/City-Wander-with-Vibe-Vision>

-Collaborative Hugging Face Dataset URL

https://huggingface.co/datasets/JocelynGUO/INFO_Shanghai_Map_Guide/discussions?status=open&sort=recently-created

-Collaborative Demo URL

<https://fionachen623.github.io/City-Wander-with-Vibe-Vision/>

-Collaborative Teaser Video URL

<https://www.capcut.cn/share/7610442542273828120?t=1>